Fundamentos de Matemática Financiera

Teoría de Carteras: ejercicio para entregar

Daniel Arrieta

darrieta@ucm.es

Universidad Complutense de Madrid

Febrero de 2026

Máster en Ingeniería Matemática

Description and Tasks

Construct and analyze efficient frontiers using monthly return data for a diverse set of assets over the period from **January 31st 2018 to October 31st 2023** (69 monthly observations).

Data Preparation

i. **Collect Monthly Return Data**: Gather monthly return data for a broad and diverse universe of financial assets, excluding the risk-free asset.

ii. **Clean and Validate the Dataset**: Address missing values and outliers to maintain data integrity and ensure reliable optimization results.

iii. **Define the Investment Universe**: Narrow down the asset pool to a manageable subset of at least 30 assets. Clearly explain the selection criteria e.g., highest average returns over the sample period or lowest average pairwise correlations to promote diversification.

Project Tasks (cont.)

Efficient Frontier Construction

Compute and visualize the efficient frontier under the following constraints (i to iii do not include a risk-free asset):

i. Unconstrained: No restrictions on asset weights (short selling allowed).

ii. Long-only: All asset weights must be non-negative.

iii. Bounded Weights: Each asset weight must lie within the interval [0, 5 %].

iv. With Risk-Free Asset and Bounded Weights: Incorporate a risk-free asset with a monthly return of $r_f = 0{,}0035$ (equivalent to an annualized rate of 4,2 %). Risky assets are subject to the same bounded weight constraint as in case (iii).

Portfolio Selection

For each efficient frontier, select one representative portfolio. Justify your selection.

Deliverables and Grading Criteria

Deliverables

A Python script implementing the analysis, and pdf report summarizing:

i. Methodology and visualizations of the efficient frontiers.
ii. Rationale for portfolio choices.
iii. Performance evaluation and conclusions.

Grading Criteria

i. Code Quality and Documentation (30 %): Clarity, organization, and comments.
ii. Methodological Rigor (30 %): Correct application of portfolio optimization techniques.
iii. Analysis and Interpretation (40 %): Depth of insights and conclusions.
iv. Out-of-Sample Performance (bonus).